

Gyan Prakash Kushwaha

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SUMMARY

AI Engineering Undergraduate (**IIT Madras**) with strong DSA fundamentals (**200+ LeetCode** problems). Experienced in building **agentic RAG workflows** and end-to-end ML systems using **LangGraph, FastAPI, and Vector DBs**. Proficient in reducing inference latency and deploying scalable AI solutions. Seeking an SDE/AI internship to leverage skills in Generative AI and backend optimization.

EDUCATION

Indian Institute of Technology (IIT), Madras

BS in Data Science and Applications; CGPA: 8.17/10

Chennai, Tamil Nadu

2023 – 2027

PROJECTS

InternOps | LangGraph, PostgreSQL, Gemini, FastAPI, Redis, Docker, Vue.js, Chrome Ext

[Repo Link](#)

- Orchestrated a **Multi-Agent AI System** using **LangGraph** and **Google Gemini** that simulates 3 distinct hiring personas, achieving **85% evaluation accuracy** and reducing application time by **95%** (from 45 mins to **<2 mins**).
- Engineered a high-performance async backend with **FastAPI**, **Celery**, and **Redis** to decouple LLM inference, implementing **semantic caching** that reduced **Time-To-First-Token (TTFT)** by **90%** and eliminated redundant API costs.
- Architected a full-stack platform combining a **Vue 3** dashboard and **Chrome Extension (Manifest V3)** with **LangSmith observability**, enabling real-time tracing of **100% of agent workflows** to debug reasoning chains and optimize system latency.

YouTube RAG Chatbot | LangGraph, FastAPI, FAISS, AsyncSQLite, Vue.js, Gemini (Flash-Lite)

[Repo Link](#)

- Reduced **Time-to-First-Token (TTFT)** by **90%** (from 5s to **<500ms**) by engineering a real-time async streaming pipeline with **FastAPI** and **LangGraph**, significantly improving user perceived responsiveness.
- Architected a **stateful RAG workflow** using **LangGraph** and **AsyncSQLite** to orchestrate persistent, multi-turn conversations, enabling context coherence across **1-hour+ video transcripts**.
- Increased RAG response accuracy by implementing **Maximal Marginal Relevance (MMR)** search, filtering **90% of redundant context** from long-form audio transcripts to minimize **LLM hallucinations**.

CropGuardian-AI | LangGraph, FastAPI, Pydantic, Gemini 1.5 Pro, OpenMeteo

[Repo Link](#)

- Architected a **multi-agent workflow** using **LangGraph** to orchestrate **3 specialized agents** (Vision, Weather, Agronomy), automating the complex reasoning chain from image analysis to actionable advice with **<3s latency**.
- Implemented strict **Structured Output** enforcement using **Pydantic** and **Gemini 1.5 Pro**, achieving **100% schema compliance** for JSON responses and eliminating parsing errors in the frontend application.
- Designed a **context-aware reasoning engine** that grounds visual diagnosis in real-time environmental data (**OpenMeteo API**), generating location-specific farming action plans based on **7-day weather forecasts**.

SKILLS

Programming Languages	Python, JavaScript (ES6+), Java, SQL
Backend Frameworks	Flask, FastAPI, Node.js
Frontend Technologies	React, Vue.js, Bootstrap, Tailwind CSS
Databases & Storage Systems	PostgreSQL, SQLite, DuckDB, Redis, FAISS, ChromaDB
Machine Learning & NLP	NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, NLTK
Generative AI & LLM Frameworks	LangChain, LangGraph, LangSmith, AGNO
DevOps & Systems	Linux, Git, GitHub, Celery
Cloud Platforms	AWS (EC2, S3, Lambda), GCP

ACHIEVEMENTS

- **Kaggle Expert**: Top 4% globally (Rank 341), 1 Silver & 9 Bronze medals; datasets with 5.6K+ downloads
- **LeetCode**: Solved 200+ DSA problems (100+ Medium)
- **HackerRank**: 5★ Gold Badge (SQL)